



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam Trimester: Summer 2025
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (10 Marks)

[CO3]

Global Descriptor Table	
Index	Address
002 H	Base: D7210111, Limit: 01234H, Access: 05H, G=1
014 H	Base: D3270011, Limit: 98765H, Access: F1H, G = 0
126 H	Base: D3741000, Limit: 1524CH, Access: 11H, G = 1
158 H	Base: D6590B00, Limit: FB000H, Access: B2H, G = 0
275 H	Base: D6870010, Limit: 0AF0EH, Access: FEH, G = 1
344 H	Base: D655000B, Limit: D015CH, Access: 07H, G = 1

An **80386 microprocessor** is executing a segment in protected mode and a portion of its descriptor table is given above. If the **index 158 H** is selected with requested privilege level **10**, then determine the followings:

a.	What was the value of the segment register ?	[3]
b.	Starting and Ending address of the segment.	[2]
c.	For an offset value of D096E H , determine the physical address.	[1]
d.	Has the segment been accessed?	[1]
e.	Segment type (CS/DS/SS/ES).	[2]
f..	What is the descriptor privilege level?	[1]

Question 2: Answer all the questions. (10 Marks)

[CO5]

a.	Device A uses an 8-bit ADC in a 10 V system, while Device B uses a 16-bit ADC . Both devices display the same ADC reading when measuring 8 V in Device A and 0.4 V in Device B . Determine the system voltage of Device B . [Show all necessary mathematical calculations.]	[4]
b.	A microcontroller-based system has two master devices : “ Master 1 ” (index: 4DH), “ Master 2 ” (index: B3H) and three slave devices : “ Slave 1 ” (index: 95H), “ Slave 2 ” (index: F5H), “ Slave 3 ” (index: 19H). Now, in I2C data transfer protocol, Master 2 receives 2-byte data (from char “ Protocol ”) from Slave 2 . Draw the corresponding sequence diagram . The ASCII values of ‘p’, ‘R’, ‘O’, ‘T’, ‘C’ and ‘L’ are respectively 112, 82, 79, 84, 67 and 76.	[4]
c.	A company is developing software for an industrial automation system where most operations involve tasks like data transfer, string handling, and arithmetic operations directly in hardware. For this design, you want to use instructions which can perform multiple operations in a single step and use a smaller number of registers. The software can be built using either RISC Architecture or CISC Architecture . Which architecture would be more suitable and why?	[2]

Question 3: Answer all the questions. (10 Marks)**[CO3]**

a.	Suppose there are eight instructions loaded into the Instruction Memory (within the Code Segment of 8086). The BIU of the 8086 will fetch these instructions for execution by the EU. The 1st instruction is a ' JUMP <5th instruction address> ', the 3rd is a ' JUMP <7th instruction address> ', the 4th instruction is a ' MOV <address> ' instruction, and the 6th instruction is a ' JUMP <2nd instruction address> ' instruction. Draw the corresponding BIU and EU cycles.	[4]
b.	Draw the timing diagram for I/O Write operation of microprocessor "808x" showing the activities of Address-Data and Address-Status pins ($AD_0 - AD_{15}$, $A_{16}/S_3 - A_{19}/S_6$), $\overline{ALE}$, $\overline{DEN}$, RD , $\overline{DT}/R$, $\overline{M}/IO$, $\overline{WR}$ in each clock cycle. [Observe the pins carefully]	[4]
c.	If there are 2048 virtual pages and each entry in the page table takes 2 Bytes, at least how many virtual pages will reside in the secondary memory given the page table size is 1 KB?	[2]

Question 4: Answer all the questions. (10 Marks)**[CO4]****RAM Content**

Address	Content	Address	Content
0H		8H	0000 0001
1H		9H	0100 1000
2H		AH	0000 1100
3H		BH	0000 0011
4H		CH	0101 0000
5H		DH	0000 0010
6H		EH	1000 0001
7H		FH	0000 0100

Opcode Table

Mnemonic	Opcode
LDA	0011
ADD	0001
SUB	0010
OUT	1101
INT	1011

Initially, the value of the Accumulator Register is 5H, the B Register is 1H, and the Output Register is 2H.

a.	Write the assembly code for the following expression using RAM content . $9 + 1 - 2 \div \frac{1}{3}$ [Division can be done using multiplication and multiplication can be done by addition]	[3]
b.	Determine the Accumulator and B register values after executing each instruction in [a].	[3]
c.	CON = $\overline{L_I} \overline{E_I} \overline{L_A} E_A E_P C_P \overline{L_M} C E E_U \overline{S_U} \overline{L_O} \overline{L_B}$ [Observe the pins carefully] While executing an instruction, the controller sequencer produces the following control words: At T₄, CON = A07, at T₅, CON = E36 and at T₆, CON = C2F. Explain what happens in each of these clock cycles and identify which operation (LDA /ADD /SUB /OUT) is being executed through these T ₄ to T ₆ .	[4]